***Solution*** Equation of the straight line, parallel to $3x-7y-12=0$ is $3x-7y+k=0$

Since it passes through the point $(6,4)$

$$\begin{aligned} 3(6)-7(4)+k &= 0 \\ k = 28-18 &= 10 \end{aligned}$$

Therefore, equation of the required straight line is $3x-7y+10=0$.

**Example 5.35** Find the equation of a straight line perpendicular to the line $y=\frac{4}{3}x-7$ and passing through the point $(7,-1)$.

***Solution*** The equation $y=\frac{4x}{3}-7$ can be written as $4x-3y-21=0$.

Equation of a straight line perpendicular to $4x-3y-21=0$ is $3x+4y+k=0$

Since it is passes through the point $(7,-1)$,

$21-4+k=0$ we get, $k=-17$

Therefore, equation of the required straight line is $3x+4y-17=0$.

**Example 5.36** Find the equation of a straight line parallel to $Y$ axis and passing through the point of intersection of the lines $4x+5y=13$ and $x-8y+9=0$.

***Solution*** Given lines $\quad 4x+5y-13=0 \quad ...(1)$

$x-8y+9=0 \quad ...(2)$

To find the point of intersection, solve equation (1) and (2)

$$\begin{array}{ccccc} x & & y & & 1 \\ 5 & \times & -13 & \times & 4 \\ -8 & & 9 & & 1 \end{array} \quad \begin{array}{ccc} & & 5 \\ \times & & \\ & & -8 \end{array}$$

$$\frac{x}{45-104} = \frac{y}{-13-36} = \frac{1}{-32-5}$$

$$\frac{x}{-59} = \frac{y}{-49} = \frac{1}{-37}$$

$$x=\frac{59}{37}, \quad y=\frac{49}{37}$$

Therefore, the point of intersection $(x,y)=\left(\frac{59}{37},\frac{49}{37}\right)$

The equation of line parallel to $Y$ axis is $x=c$.

It passes through $(x,y)=\left(\frac{59}{37},\frac{49}{37}\right)$. Therefore, $c=\frac{59}{37}$.

The equation of the line is $x=\frac{59}{37} \Rightarrow 37x-59=0$

**Example 5.37** The line joining the points $A(0,5)$ and $B(4,1)$ is a tangent to a circle whose centre $C$ is at the point $(4,4)$ find

(i) the equation of the line $AB$.

(ii) the equation of the line through $C$ which is perpendicular to the line $AB$.

(iii) the coordinates of the point of contact of tangent line $AB$ with the circle.



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